CSL301 : Operating System
LAB-3
Marks-100

Q1. Implement head command 25 marks

Implement a simplified version of the Linux head command using low-level file handling.

Your program should:

- Accept a filename as a command-line argument.
- Open the file using open().
- Read the file using read().
- Display only the first 5 lines of the file.
- Use write() to display the output.
- Close the file using close().

Example:

./head5 sample.txt

Expected behavior:

Line 1
Line 2
Line 3
Line 4
Line 5

Restriction: Do not use fopen(), fgets(), or other high-level file functions.

Q2. Implement cp command 25 marks

Implement a simplified version of the Linux cp command using system calls.

The program should:

./mycp source.txt destination.txt

Copy all contents of source.txt into destination.txt.

Requirements:

1. Open the source file using O_RDONLY.
2. Create/open the destination file using O_CREAT | O_WRONLY | O_TRUNC.

3. Use read() to read data from the source.
4. Use write() to write data to the destination.
5. Continue until the complete file is copied.
6. Close both file descriptors.
7. Display an error if either file cannot be opened.

Q3. Implement grep with line numbers 25 marks

The program should accept a **search word** and a **filename**:

./mygrep error log.txt

It should display every line containing the search word along with its line number.

Example file:

System started

User logged in

Error opening file

Connection successful

Error reading file

Output:

3: Error opening file

5: Error reading file

Requirements:

- Use open() and read().
- Process the file without loading the entire file into memory.
- Keep track of the current line number.
- Search for the given word.
- Display matching lines using write() or printf().

Q4. Implement a mini fileinfo command 25 marks

Create a program called fileinfo that accepts a filename:

./fileinfo sample.txt

The program should display:

File: sample.txt

Lines: 10

Words: 45
Bytes: 238

In addition, display whether the file is:

Readable: Yes
Writable: Yes

Use appropriate system calls/library functions to determine the file's properties.

Submission Checklist

- Terminal output / screenshots supporting the execution
- C files
- Include everything in a single zip file.

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